

Dynamics: Newton laws

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Three passengers are in a car that broke down. One of the passengers decided to push the car

Is he able to crank the car to get it started?

No, what to do ?

Ask the other passenger to help. Therefore,

- 1 The total weight of the car and passengers: **object** is reduced.
- 2 Two passengers increase the strength: **Force**

The force is an action which changes the state or the form of an object (possibly both).

The state of an object is depending on : the object mass and the force used on the object.

We can exert a force and nothing happens, as pushing a wall with hands.

Force Types

For two objects (A) et (B), the force $\vec{F}_{BA}$ exerted by B on A , is said to be:

- **external** if B is considered as a system apart.
- **internal** ($A + B$) form one system.

Contact Force: When it involves direct contact between two bodies, such as push a car. A force is a vector with magnitude and direction.

Non-contact Force : Some object can interact without a direct contact,

Example : Earth-Moon interaction

This interaction occurs due to the gravitational field (a concept introduced by Faraday).

Field Forces

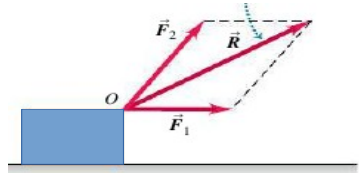
characterize interacting objects, which do not have physical contact.

Forces Superposition

The magnitude is measured by a scaled spring with Newton (N) as SI unit. When more than one force is applied on an object: the total force (**Resultant**) is obtained by adding vectors.



Spring scale

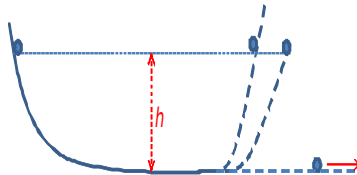


Adding two forces: $\vec{R} = \vec{F}_1 + \vec{F}_2$

when many forces (n) are applied at an object, they have the same effect as a single force equal to the vector sum of the forces: $\vec{R} = \sum_i^n \vec{F}_i$.

Newton's First law

A ball is released from a height of h and rolls down an inclined plane. Another ball moves in the opposite direction and meets the first ball at the same height of h , regardless of the angle of the inclined plane. This is true if there is no friction.



Galilée principale

An isolated object has a constant speed (at rest if $v = 0$ in the frame of reference).

Newton's First Law of Motion: A body acted on by no net force has a constant velocity.

Inertial frame of Reference

An isolated object has a constant velocity: straight-line motion.

Any frame in which the object acceleration is null, this frame is inertial.

Example 1

If two trains are moving with constant velocities (vector) to someone who is in the train stations, both are inertial frame of reference even if their velocities are different.

Question 1

If one inertial frame is know, we can find an infinite frame of references, why?.

Any object moving with constant vector velocity in this frame is also an inertial frame.

Let A be an inertial frame of reference. Any second frame of reference B will also be inertial if it moves relative to A with constant velocity $\vec{v}$. For any mobile M moving with constant velocity relative to A , then:

$$\vec{v}_{M/A} = \vec{v}_{M/B} + \vec{v}_{B/A}$$

- For two balls, one metallic (bowling) and one soft (tennis), moving on a smooth surface, which one moves faster if we apply the same force?

The tennis ball because the **mass** is smaller:

- The bowling ball of mass m_2 strikes the tennis ball, of mass m_1 . Both balls' velocities change, but the change in velocity is more significant for the tennis ball.

$$v_1'(\text{after the shock}) > v_1(\text{before the shock})$$

- To stop an object of mass m and velocity $\vec{v}$, we must apply a force which depends on m and $\vec{v}$. Therefore, we define the **momentum** or linear momentum by

$$\vec{p} = m\vec{v}, \quad \vec{p} \parallel \vec{v}$$

of unit kg.m/s

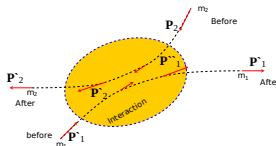
Momentum

Can be **added**: when two or more bodies that interact.

System	m_1	m_2	$(m_1 + m_2)$
Momentum	$\vec{p}_1$	$\vec{p}_2$	$\vec{p}_T = \vec{p}_1 + \vec{p}_2$

Conservation

For an isolated system the total momentum is constant $\vec{p} = \overrightarrow{const}$



Phase / Interaction	before	during	after
Isolated system	m_1, m_2	$(m_1 + m_2)$	m_1, m_2
Momentum m_1	$\vec{p}_1 = \overrightarrow{const}$	$\vec{p}_1' = \text{variable}$	$\vec{p}_1' = \overrightarrow{const}$
Momentum m_2	$\vec{p}_2 = \overrightarrow{const}$	$\vec{p}_2' = \text{variable}$	$\vec{p}_2' = \overrightarrow{const}$
Momentum $(m_1 + m_2)$	$\vec{p}_T = \vec{p}_1 + \vec{p}_2 = \vec{p}_1' + \vec{p}_2' = \vec{p}_1' + \vec{p}_2' = \overrightarrow{const}$		

Example 2

A lorry of mass $M = 2 \text{ t}$ hits a car of mass $m = 1000 \text{ kg}$ moving with the velocity ($V = 80 \text{ km/h}$) perpendicular to the lorry velocity. After the shock the lorry and the car remain stick to forme one system of an angel $\frac{\pi}{4}$ to the initial lorry direction. A witness says that the car was moving with a speed greater than 80 km/h , did he tell the truth?

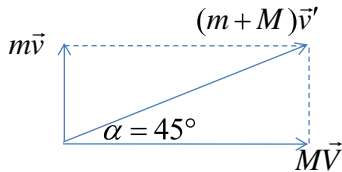
Answer:

Before the shock : The car momentum is $m\vec{v}$ and the lorry momentum is $M\vec{V}$. After we have (Lorry+ car) with $(m + M) \vec{v}'$ of an angle $\frac{\pi}{4}$ with respect to $M\vec{V}$. As the system is isolated then, $m\vec{v} + M\vec{V} = (m + M) \vec{v}'$.

From

the figure we can show,

$\frac{mv}{MV} = \tan(\alpha) = 1 \Rightarrow v = \frac{MV}{m} = \frac{2000 \times 80}{1000} = 160 \text{ km/h}$. The witness is not a liar.



Newton's Second Law

When an ice cube slides on a smooth surface: **no friction**, by acting a force $\vec{F}$, the object acceleration is $\vec{a}$.

For two blocks to have the same acceleration $\vec{a}$ the force must be : $2\vec{F}$

Newton's 2nd law

In an inertial frame of reference, the acceleration of an object is proportional to the net force and to the inverse of the mass.

$$\sum \vec{F} = m\vec{a}$$

However, when the mass is not constant then,

$$\vec{F} = \frac{d\vec{p}}{dt} = m \frac{d\vec{v}}{dt} + \frac{dm}{dt} \vec{v}$$

The force unit Newton (N) means that when an object of mass of 1 *kg* is under a force of 1 *N*, the object acceleration is 1 *m/s*²

- If the acceleration on an object is equal to zero what does it mean?
No net force acting on the object.
- The forces resultant leads an object to acquire an acceleration opposite to the force direction. is it true, why?
No, this is not true because the acceleration vector is proportional to the force (mass > 0).
- A force $\vec{F}$ is applied on an object during an interval of time ΔT , how does the velocity change?

$$F = \frac{m\Delta v}{\Delta t}, \quad \Delta v = F\Delta t/m$$

Example: Atwood machine

Two objects of unequal mass are hung vertically over a frictionless pulley of negligible mass. Determine the magnitude of the acceleration of the two objects and the tension in the lightweight cord.

How this machine can be used?

Apply Newton's law to (1) and (2):

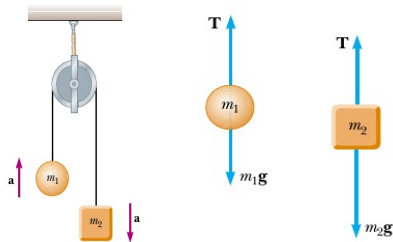
$T_1 - m_1g = m_1a_1$, $m_2g - T_2 = m_2a_2$ The string is inextensible:

$a_1 = a_2 = a$.

The pulley is of negligible mass: $T_1 = T_2 = T$ Then,

$$a = \frac{(m_2 - m_1)}{(m_2 + m_1)}g$$

If we let m_2 close to m_1 then we can measure a to find g .



Newton's third law

An isolated system contains two objects. Object (1) exerts on (2) a force $\vec{F}_{12}$. What would be the response of object (2).

As the system is isolated: $\vec{p} = \vec{p}_1 + \vec{p}_2 = \vec{C}$ (constant). Then,

$$\begin{aligned}\frac{d\vec{p}}{dt} &= \frac{d\vec{p}_1}{dt} + \frac{d\vec{p}_2}{dt} = \vec{0} \\ \frac{d\vec{p}_1}{dt} &= -\frac{d\vec{p}_2}{dt} \\ \vec{F}_{21} &= -\vec{F}_{12}\end{aligned}$$

Action-reaction

The action (force) exerted by the object (1) on object (2) is equal and opposite to the reaction (force) of (2) on (1).

Gravitational Force

The Earth attracts any object by the gravitational $\vec{F}_g$.

When objects of masses $m_1, m_2, \dots$ are dropped at the same place, their accelerations verify

$$a_1 = a_2 = \dots = 9.81 \text{ m/s}^2 \equiv g$$

From Newton's second law, $\vec{F} = m\vec{a}$, but $\vec{a} = \vec{g}$. The object's weight is $\vec{F} \equiv \vec{P}, P$

$$\vec{P} = m\vec{g}$$

where the gravitational acceleration is $\vec{g} = -g\vec{k}$. $\vec{k}$ is an up vertical vector and $g = 9.81 \text{ m/s}^2$.

Close to the Earth's surface the gravitational force is known as **weight**

We can also apply action-reaction principal to say that the object exerts a force on the Earth. Therefore,

Between tow objects a **gravitational force** exists.

Example 3

The Moon motion
relative to the Earth is uniform circular:

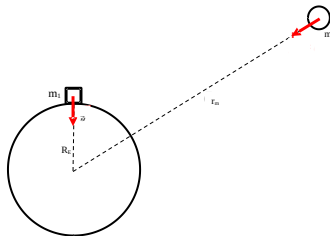
$$v = \omega R, \quad a_t = 0, \quad a_n = \omega^2 R$$

The gravitational force on the moon
 F corresponds to its weight relative to Earth.

We know:

$g = 9.81 m/s^2$, $R_E = 6400 km$ (Earth radius),
the Earth-Moon distance is $r_M = 60R_E$,
and the period is $T = (27 + \frac{1}{3})$ days.

Earth-Moon



1. Find a_M :

$$a_M = \omega^2 r_M = \left(\frac{2\pi}{T}\right)^2 r_M = \frac{4\pi^2 r_M}{T^2} = \frac{4\pi^2 \times 60 \times 6400 \times 10^3}{(27 \times 24 \times 3600 + \frac{24}{3} \times 3600)^2} = 2.72 \times 10^{-3} \text{ m/s}^2$$

2. We compare between P and F by using (m, m_M, r_M, R_E)

$$\frac{P}{F} = \frac{m_1 g}{m_M a_M} = \frac{m_1}{m_M} \frac{9.81}{2.72 \times 10^{-3}} = 3606.6 \frac{m_1}{m_M} \approx 3600 \frac{m_1}{m_M} = \frac{r_M^2}{R_E^2} \frac{m_1}{m_M} \Rightarrow F = K_E \frac{m_M}{r_M^2} \text{ et } P = K_E \frac{m_1}{R_E^2}.$$

E refers to Earth and K_E is the gravitational constant relative to the force exerted by the Earth on Moon. P and F are exerted by Earth. Similarly, the force exerted by Moon on Earth is

$$F' = K_M \frac{M_E}{r_M^2}$$

where M_E is the Earth's mass.

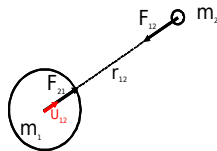
The Gravitational Force

Every body of mass m_1 attracts every other body of mass m_2 with a force that is directly proportional to the product of the masses of the particles and inversely proportional to the square of the distance between them.

$$\vec{F}_{12} = -G \frac{m_1 m_2}{\|\vec{r}_{12}\|^2} \vec{u}_{12}, \quad \text{the unit vector } \vec{u}_{12} = \frac{\vec{r}_{12}}{\|\vec{r}_{12}\|} \text{ goes from 1 to 2}$$

where $G = 0.667 \times 10^{-10} \text{ m}^3/\text{kg} \cdot \text{s}^2$
is the gravitational constant and $\vec{F}_{12}$ is the
force exerted by 1 on 2. where $\vec{r}_{12} = \vec{r}_2 - \vec{r}_1$.
Similarly the force of 2 on 1 is opposite,

$$\vec{F}_{12} = m_2 \vec{g}_{12}(r) \quad \text{where} \quad \vec{g}_{12}(r) = -G \frac{m_1}{r_{12}^2} \vec{u}_{12}$$



$\vec{g}_{12}(r)$ is the field produced by m_1 at the point 2.

There is a field associated with each force that acts at a distance.

Weight

In the case of an object of mass m located at an altitude h from the Earth's surface, the weight is

$$P = m \cdot G \frac{m_E}{r^2}$$

$r = R_E + h$: (from the the center of the earth to the object), which yields

$$P = m \cdot G \frac{m_E}{(R_E + h)^2}$$

For $m = 1 \text{ kg}$ and $h = 100$, one finds

$$P_1 = 1 \times 0.67 \times 10^{-10} \frac{5.972 \times 10^{24}}{(6400 \times 10^3 + 100)^2} = 9.7683 \text{ N}$$

and for $h = 10000 \text{ m}$

$$P_2 = 1 \times 0.67 \times 10^{-10} \frac{5.972 \times 10^{24}}{(6400 \times 10^3 + 10000)^2} = 9.7382 \text{ N}$$

as we know the weight corresponds to $P = mg = 9.8 \text{ N}$, considered constant if $h \ll R_T$.

For two planets of masses respectively m_1 et m_2 of our solar system their circular motion radius are R_1 et R_2 of periods T_1 et T_2 .

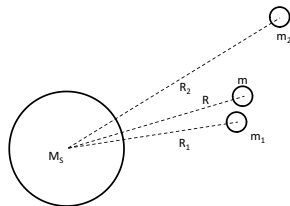
From the gravitational law we obtain,

$$\begin{cases} F = ma = m \frac{4\pi^2 R}{T^2} \\ F = G \frac{M_s m}{R^2} \end{cases} \Rightarrow \frac{R^3}{T^2} = G \frac{M_s}{4\pi^2} = C^{te}$$

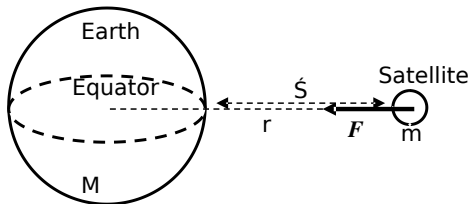
where M_s is the solar mass

The Kepler's law of planet periods is

$$\frac{R_1^3}{T_1^2} = \frac{R_2^3}{T_2^2} \Rightarrow \left(\frac{T_1}{T_2} \right)^2 = \left(\frac{R_1}{R_2} \right)^3$$



For satellite with the same period as Earth, located at a specific altitude and longitude, the period is $T=24$ h



From Newton's law: $F = G \frac{Mm}{r^2} = ma = m\omega^2 r$.

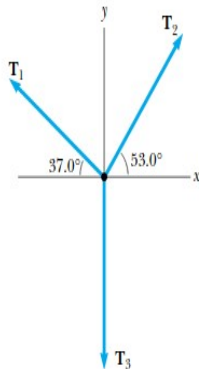
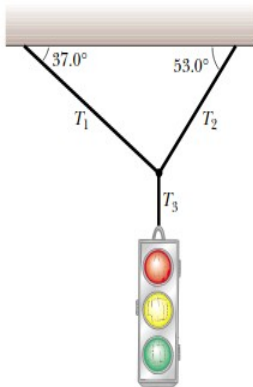
What is the satellite position?

One finds $r = \left(\frac{GM}{\omega^2}\right)^{1/3} = \left(GM \frac{T^2}{4\pi^2}\right)^{1/3}$. as $M = 5.97 \times 10^{24} \text{ kg}$, then $r = 42200 \text{ km}$ and $h = r - R_T = 35800 \text{ km}$

The satellite speed is $v = \omega r = (GM\omega)^{1/3} = \left(GM \frac{2\pi}{T}\right)^{1/3} = 3.07 \text{ km/s}$.

Equilibrium

A traffic light of 122 N weight is hunged by vertical strings (Figure). The latter make angles of 37.0° and 53.0° with the horizontal line, respectively. The tension on the inclined strings is limited to 100 N . Can this object remains hanged?



Equilibrium: $\sum_{\text{ext}} \vec{F} = \vec{0}$, $\vec{P} + \vec{T}_3 = \vec{0}$, $P - T_3 = 0$, So $T_3 = P = 122 \text{ N}$

But: $\vec{T}_3 = \vec{T}_1 + \vec{T}_2$ by projection on vertical and horizontal axis:

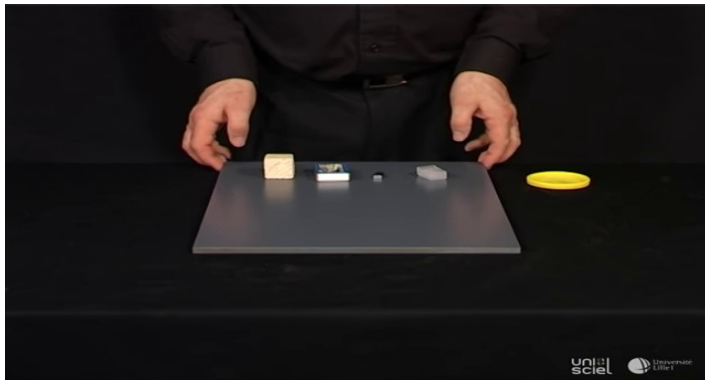
$$\sum F_x = -T_1 \cos(37.0) + T_2 \cos(53.0) = 0 \quad (1)$$

$$\sum F_y = T_1 \sin(37.0) + T_2 \sin(53.0) - 122 = 0 \quad (2)$$

One finds: $T_1 = 73.4 \text{ N}$ et $T_2 = 97.4 \text{ N}$

As the tension is limited to 100 N for each string, the traffic-light remains hanged.

Put on a horizontal table different objects (three). *click on the the image:*



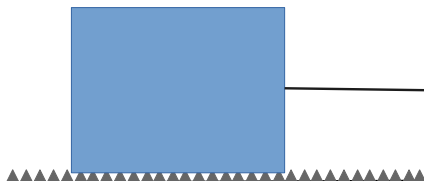
What are the main conclusion?

When an object is on surface, we must apply a minimum force to make it moving

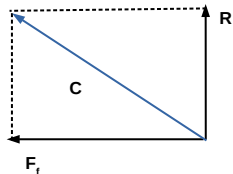
From the previous example with different type of object (wood, glass and paper):

The motion depends
on the **contact** surface nature.

Let us write the newton equation
of a body of mass m under the
action of such a minimum force:



- $\vec{F}$: the acting Force .
- $\vec{P}$: object's weight.
- $\vec{R}$: surface force on the object .
- $\vec{F}_f$: Friction force, acting against the object motion under the effect of $\vec{F}$.



The force $\vec{F}$ can't lead the object to move.

There is a force which "resist" to the object motion said **static friction force $\vec{F}_f$** .

The resultant forces $\vec{C}$ is: $\vec{C} = \vec{R} + \vec{F}_f$: Contact force.

The object starts to move when $F > F_s$.

This depends only on the contact surface nature

Contact Forces

For two bodies in contact the force $\vec{C}_{21}$ of (2) on (1) is equal and opposite to $\vec{C}_{12}$ the force of (1) on (2): $\vec{C}_{12} = -\vec{C}_{21}$.

$\vec{C}$ has no general mathematical expression, **calculated by using Newton's laws.**

When the object starts moving , it can be kept moving with less force needed to get it started. The force opposite to motion is said : **kinetic friction force.**

The contact fore has to components:

- $\vec{C}_{\parallel}$: parallel to the contact surface. If the object slides , $\vec{C}_{\parallel} \equiv \vec{f}$: kinetic friction force has a sens of $(-\vec{v})$.
At rest (relative to the surface) it is called static friction force.
- $\vec{C}_{\perp}$: perpendicular to the contact surface.

Friction Coefficients

The static friction coefficients μ_S is the limit for the object to slide μ_k characterize the object when there is motion (the surface is not smooth).

$$\mu_S = \frac{C_{0\parallel}}{C_{0\perp}}, \quad \mu_k = \frac{C_{\parallel}}{C_{\perp}}$$

Depend on contact surface matter.

Properties:

- μ_S and μ_k are independent from velocity.
- μ_S and μ_k are independent from the size of contact surfaces.
- μ_S and μ_k depend on the contact surface composition. For the same nature , $\|\vec{C}_{0\parallel}\|, \|\vec{C}_{\parallel}\|, \|\vec{C}_{0\perp}\|$ and $\|\vec{C}_{\perp}\|$ may change with masses but μ_S and μ_k remain constants.
- $\mu_k < \mu_S$.
- μ_S et μ_k characterize the contact.

Measuring friction coefficients

We use the dynamometer.

With the force $\vec{F}$ which increases gradually,
The object starts to move at a critical value F_r

$$\vec{F} = \vec{F}_s$$

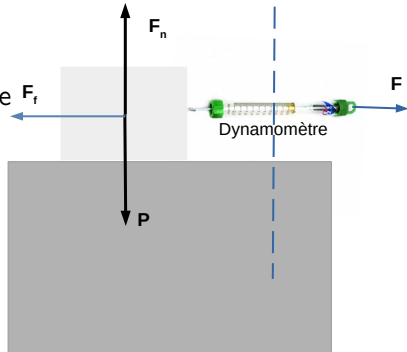
The friction force (F_s) is depicted from
the dynamometer .

At equilibrium: :

$$F_n = P \quad F_f = F_s$$

We find,

$$\mu_s = \frac{F_f}{F_n} = \frac{F}{P}$$



Example: Inclined Plane

θ_s : critical angle

corresponds to equilibrium break-down.

Forces at equilibrium :

$$\text{ox: } f_s = mg \sin(\theta_s)$$

$$\text{oy: } N' = mg \cos(\theta_s)$$

For $\theta > \theta_s$ the mobile acceleration is a

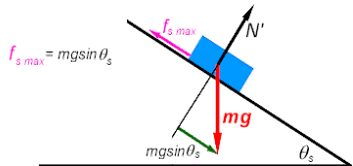
$$\text{ox: } f_f = m(g \sin(\theta) - a)$$

$$\text{oy: } N' = mg \cos(\theta)$$

Kinetic friction coefficient:

$$\mu_k = \frac{g \sin(\theta) - a}{g \cos(\theta)}$$

Lorsque le bloc est sur le point de se mettre en mouvement ...



Elastic Force

The elastic force is the force that arises from the deformation of a solid such as a spring or a rubber band.

An ideal **spring** of length l_0 and a spring constant k (N/m).

Can be compressed ($l < l_0$) or stretched ($l > l_0$).

The force applied on an object suspended at the free end of the spring is a **restoring or elastic** force which is: proportional and opposite to $\Delta \vec{l}$:

$$\vec{T} = -k\Delta \vec{l}, \quad \text{Hook's law: } \vec{F} = -k\vec{x}$$

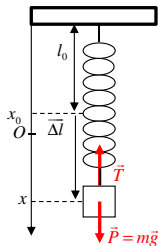


Fig. 1

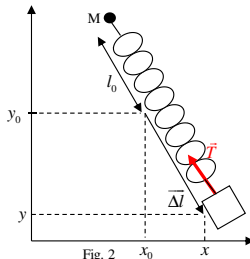
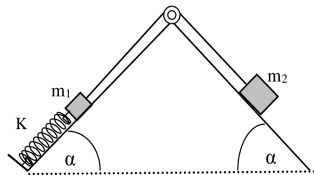


Fig. 2

Two bodies of masses $m_1 = 1 \text{ kg}$ and m_2 are linked by inextensible string through a pulley of negligible mass. The objects are put on inclined plane (Figure), where $\alpha = 45^\circ$. The spring is of constant $k = 100 \text{ N/m}$. The contact between the surface and the bodies is characterized by a static and dynamic friction coefficients $\mu_s = 0.5$ and $\mu_d = 0.3$.

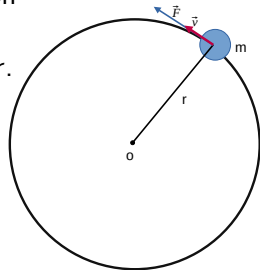
1. The spring is free, find the mass m_2 corresponding to equilibrium break-down.
2. Let $m_2 = 5 \text{ kg}$, show the forces applied on the two bodies when the spring is stretched by 1 cm .
3. Find the acceleration of the system.
4. Find the spring tension.



To study translational motion we use Newton's equation second law,

$$\sum_{ext} \vec{F} = m\vec{a}$$

Let us consider the rotation of a point-like object of mass m and position r to the axis of rotation O . The rotation is due to a force $\vec{F}$ which, for a simplicity is considered perpendicular to r . parallel to $\vec{v}$



$$F = m \frac{dv}{dt} \quad \text{but} \quad v = r \cdot \dot{\theta}$$

as the radius is constant,

$$F = mr \frac{d\dot{\theta}}{dt} \quad \text{or} \quad F \cdot r = mr^2 \frac{d\dot{\theta}}{dt}$$

If we want this equation to look like Newton's equation we set,

Moment of inertia

$F.r = \tau$ torque (N. m) and $m.r^2 = I$ moment of inertia ($kg.m^2$)

$$\tau = I.\ddot{\theta}$$

r is called **lever arm**

When

the force is not perpendicular to the lever arm, we write $\vec{F}$ in term of two components $\vec{F} = \vec{F}_{\parallel} + \vec{F}_{\perp}$.

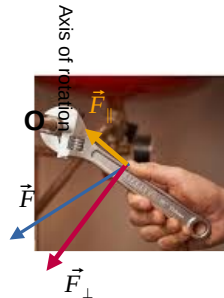
$\vec{F}_{\parallel}$ have no effect on the the rotational motion (responsible of translation).

we can write

$$\tau = F_{\perp}.r = I\ddot{\theta} \quad ; \quad F.r.\sin(\theta) = I\ddot{\theta}$$

θ being

the angle between the lever arm and the force.



In vectors representation we write

$$\vec{\tau} = \vec{r} \wedge \vec{F}$$

The torque τ is a vector $\perp$ to the plane $\vec{r}, \vec{F}$
when there is more than one torque the second Newton's law for rotation corresponds to,

$$\sum \tau = I \cdot \ddot{\theta}$$

For a rigid body we apply this relation to small masses $m_1, m_2, \dots m_i$ of radius $r_1, r_2, \dots r_i$, respectively that are measured from the axis of rotation chosen to be the axis of **symmetry**.

$$\sum \tau = m_1 r_1^2 \cdot \ddot{\theta}_1 + m_2 r_2^2 \cdot \ddot{\theta}_2 + \dots + m_i r_i^2 \cdot \ddot{\theta}_i$$

but all points of the **rigid** body undergo the same rotation,

$$\theta_1 = \theta_2 = \dots = \theta_i \quad \text{so} \quad \ddot{\theta}_1 = \ddot{\theta}_2 = \dots = \ddot{\theta}_i = \ddot{\theta}$$

Rigid body

The Newton's law for the rotation of a rigid body

$$\sum \tau = \sum_1^N (m_i r_i^2) \ddot{\theta} = I \ddot{\theta}$$

The moment of inertia is

$$I = \sum (m_i r_i^2)$$

when body has a continuous mass distribution then

$$I = \int_{body} r_i^2 dm$$

Moment of inertia

of symmetric objects about the center (of symmetry),

- Point-like objects : $I = mr^2$
- Hoop of mass M and radius R :
 $= MR^2$
- Cylinder or disc

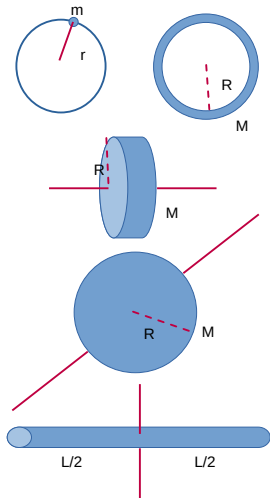
$$I = \frac{Mr^2}{2}$$

- Sphere

$$I = \frac{2}{5}Mr^2$$

- Rode of length L about center

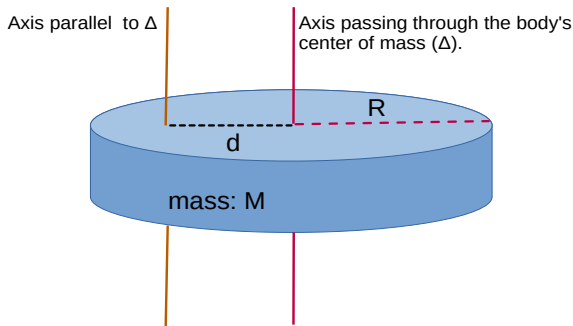
$$I = \frac{ML^2}{12}$$



Huygens–Steiner theorem

When the axis of rotation is different from the axis of symmetry but remaining parallel at a distance d , the new moment of inertia is

$$I = I_{\Delta} + Md^2$$



Exercise 1

Find the acceleration of a sphere of radius $R = 10 \text{ cm}$ and mass $m = 1 \text{ Kg}$ when it rolls without slipping on an inclined plane about an angle of $\alpha = 30^\circ$ to the horizontal. The kinematic friction coefficient is $\mu_k = 0.3$

The translation motion is governed by,

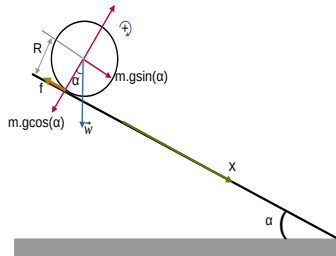
$$\sum \vec{F} + \vec{C} = m\vec{a}; \quad mgsin(\alpha) - f = ma_x;$$

The rotational motion is governed by,

$$\vec{\tau} = I\vec{\theta}; \quad f.R = I\frac{a_x}{R}$$

The acceleration along the inclined plane is

$$a_x = \frac{m.sin(\alpha)}{(m + I/R^2)}g = \frac{m.sin(\alpha)}{(m + 2m/5)}g = \frac{5}{7}sin(\alpha)g$$



What happens when $f = 0$?

Exercise 2

Find the acceleration in Atwood machine whe the pulley is a disc of mass m and radius R

For translation of masses (1) and (2):

$$T_1 - m_1g = m_1a_1, \quad m_2g - T - 2 = m_2a_2$$

The string is inextensible: $a_1 = a_2 = a$.

The pulley is of negligible mass:

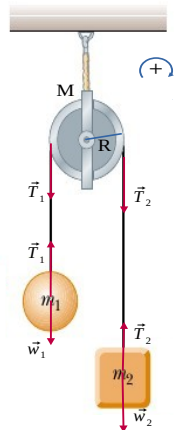
$$T_1 = T_2 = T$$

for the rotation of the pulley

$$\sum_i \tau_i = I \ddot{\theta}, \quad T_2 R - T_1 R = I(a/R)$$

Then ($I = MR^2$),

$$a = \frac{(m_2 - m_1)}{(m_2 + m_1 + I/R^2)}g = \frac{(m_2 - m_1)}{(m_2 + m_1 + M)}g$$



Angular momentum

In translation motion Newton's equation can be written as,

$$\sum_{\text{ext}} \vec{F} = \frac{d\vec{p}}{dt}$$

To obtain a similar equation for rotational motion we set,

$$\vec{L} = \vec{r} \wedge \vec{P} \quad \text{Angular momentum}$$

Derivation to time gives,

$$\frac{d\vec{L}}{dt} = \frac{d\vec{r}}{dt} \wedge m\vec{v} + \vec{r} \wedge m \frac{d\vec{v}}{dt}$$

as

$$\vec{v} = \frac{d\vec{r}}{dt}, \quad \text{and} \quad \vec{v} \wedge \vec{v} = \vec{0}$$

One obtains

$$\frac{d\vec{L}}{dt} = \vec{r} \wedge \vec{F} = \sum_i \vec{\tau}_i$$

Conservation

For a rigid body

$$L = \sum_i r_i \cdot p_i = \sum_i L_i = \sum_i r_i m_i v_i = \sum_i r_i m_i r_i \omega = \left(\sum_i m_i r_i^2 \right) \omega$$

$$L = I \cdot \omega$$

Conservation

For an isolated system the angular momentum is conserved

$$\frac{dL}{dt} = 0$$

For two rotating systems of moment of inertia I_1 , I_2 and angular velocities ω_1 , ω_2 , respectively, when combined to form one system of moment of inertia $I_1 + I_2$, the angular velocity of this system is ω ,

$$I_1 \omega_1 + I_2 \omega_2 = (I_1 + I_2) \omega; \quad \omega = \frac{I_1 \omega_1 + I_2 \omega_2}{I_1 + I_2}$$